

Software Engineering - Question Bank (Unit-wise)

BCA-204 (4th Sem): Jul 2023, Jun 2024, May-Jun 2025 | BCA-108T (2nd Sem): May-Jun 2025

Every question from the four papers is arranged under the unit and syllabus topic it belongs to, using the syllabus in your screenshot. Sub-parts are listed separately, so each row is one answerable question.

In syllabus	Topic is clearly listed in the syllabus.
PARTIAL	Related to a listed topic, but the exact term or part is not named.
NOT IN SYLLABUS	Not covered by the syllabus (shaded red in the tables).

At a glance

Unit	Questions	Partial	Not in syllabus
UNIT I	23	2	1
UNIT II	18	1	3
UNIT III	21	0	3
UNIT IV	19	2	0
Total	81	5	7

Source key: the Source column gives paper and question number. Marks are as printed; in Jun-25 (BCA-204) Q1 each part carries 4 marks. A * means marks are shared with the other parts of that question (Jun-24 Q7 has three short notes for 12.5 marks in all).

Note: The Jun-25 BCA-108T paper is a 2nd-semester paper, but its questions were mapped to the same syllabus you supplied.

Questions outside the syllabus (quick list)

Question	Source	Unit
What are the characteristics of software?	Jul-23 (BCA-204) Q1(a)	UNIT I
Discuss the need of data structure metrics.	Jun-24 (BCA-204) Q1(c)	UNIT II
Compare the Walston-Felix and SEL models for software involving 8 person-years effort: (i) LOC produced (ii) duration (iii) productivity in LOC/PY (iv) average manning.	Jun-24 (BCA-204) Q5	UNIT II
Explain the Walston & Felix model and compare with the SEL model.	Jun-25 (BCA-204) Q4(a)	UNIT II
What is design? Describe the difference between conceptual design and technical design.	Jun-25 (BCA-204) Q6(b)	UNIT III
Define: Objects, Message, Abstraction, Class, Inheritance and Polymorphism.	Jun-25 (BCA-204) Q7(b)	UNIT III
What is CMM? How is it different from ISO 9000?	Jun-25 (BCA-108T) Q7(b)	UNIT III

UNIT I

Syllabus: Intro to SE: software crisis; life cycle models - Waterfall, Prototype, Spiral, Agile. Requirements analysis & specification: requirement engineering, elicitation (FAST, QFD), analysis using DFD / use-case / sequence / class diagrams, ER diagrams, SRS - characteristics & organisation.

Introduction, Software Crisis & SDLC Models

#	Question	Source	Marks	Status
1	What is software crisis? Was Y2K a software crisis?	Jun-25 (BCA-204) Q1(a)	4	In syllabus
2	Why are we in perpetual software crises? Explain.	Jun-25 (BCA-108T) Q1(g)	5	In syllabus
3	What is software engineering? Is it an art, craft or a science? Discuss.	Jun-25 (BCA-204) Q1(b)	4	In syllabus
4	Define the term "Software engineering". Explain the major differences between software engineering and other traditional engineering disciplines.	Jun-25 (BCA-204) Q2(a)	5	In syllabus
5	What is System Development Life Cycle (SDLC)? Explain its phases.	Jun-25 (BCA-204) Q1(e)	4	In syllabus
6	What are the major phases in the Waterfall model and Spiral model? Where is the Spiral model beneficial?	Jun-25 (BCA-204) Q2(b)	5	In syllabus
7	Explain Agile Model with its advantages and disadvantages.	Jul-23 (BCA-204) Q2(b)	6.5	In syllabus
8	Explain Agile Model in detail. What are its advantages and disadvantages? Compare and contrast with Waterfall, Prototype and Spiral SDLC models.	Jun-25 (BCA-108T) Q3	10	In syllabus
9	Discuss the RAD and Prototype SDLC models. <i>Note: Prototype is in the syllabus; RAD is not listed.</i>	Jun-24 (BCA-204) Q1(g)	5	PARTIAL
10	What are the characteristics of software? <i>Note: Only 'Software crisis' is listed under Introduction; characteristics of software are not.</i>	Jul-23 (BCA-204) Q1(a)	5	NOT IN SYLLABUS

Requirements Analysis & Specification

#	Question	Source	Marks	Status
1	What do you mean by requirements elicitation? Discuss in brief different requirements elicitation techniques.	Jul-23 (BCA-204) Q2(a)	6	In syllabus
2	Explain the various requirement elicitation techniques in detail.	Jun-24 (BCA-204) Q2	12.5	In syllabus
3	What are the various requirement elicitation methods? Discuss FAST in brief.	Jun-25 (BCA-108T) Q1(a)	5	In syllabus
4	Define FAST and QFD.	Jun-25 (BCA-204) Q1(h)	4	In syllabus
5	Write short notes on DFD and ER diagrams.	Jul-23 (BCA-204) Q1(c)	5	In syllabus
6	Draw the E-R diagram, Use case diagram and Two-level DFD for Library Management System.	Jun-24 (BCA-204) Q3	12.5	In syllabus
7	Draw and label a well-described Use Case Diagram and Level-1 DFD for a hotel management system. Make assumptions as required.	Jul-23 (BCA-204) Q3(a)	6	In syllabus
8	Draw a DFD for borrowing a book in a library: "A borrower can borrow a book if it is available, else he/she can reserve for the book if he/she so wishes. He/she can borrow a maximum of three books."	Jun-25 (BCA-204) Q3(a)	5	In syllabus

#	Question	Source	Marks	Status
9	Movie-Time Cinemas ticket reservation system (theatres, stages, numbered seats, showings; information received automatically). Design the following, writing assumptions: (a) Use case diagram (b) Class diagram.	Jun-25 (BCA-108T) Q2	10	In syllabus
10	Write a short note on Data dictionary. <i>Note: Not listed by name; it is the companion of DFD analysis.</i>	Jun-25 (BCA-204) Q1(d)	4	PARTIAL
11	What is SRS? Describe the nature and characteristics of SRS. Why is it important?	Jul-23 (BCA-204) Q3(b)	6.5	In syllabus
12	What are the characteristics of a good SRS?	Jun-24 (BCA-204) Q1(e)	5	In syllabus
13	What is Software Requirements Specification (SRS)? List out the advantages of SRS standards. List five desirable characteristics of a good SRS document.	Jun-25 (BCA-204) Q3(b)	5	In syllabus

UNIT II

Syllabus: Project planning: software metrics (definition, need, Product / Process / Project metrics); size estimation (LOC, Function Count, Halstead); cost estimation (need, COCOMO basic & intermediate). Risk management: risks, types, risk assessment, risk control.

Software Metrics

#	Question	Source	Marks	Status
1	What is Software Metrics? Explain various types of metrics - Product, Process and Project metrics in brief.	Jun-25 (BCA-108T) Q5	10	In syllabus
2	Define Deliverables, Milestones, Metric, Measure and Measurement. <i>Note: Metric / measure fall under 'Definition of metrics'; deliverables and milestones are not listed.</i>	Jun-25 (BCA-108T) Q1(c)	5	PARTIAL
3	Discuss the need of data structure metrics. <i>Note: Syllabus lists only Product, Process and Project metrics; data structure metrics are not covered.</i>	Jun-24 (BCA-204) Q1(c)	5	NOT IN SYLLABUS

Size Estimation (LOC, Function Count, Halstead)

#	Question	Source	Marks	Status
1	Describe any two software size estimation techniques.	Jun-24 (BCA-204) Q1(b)	5	In syllabus
2	What are the various activities during software project planning? Describe any two software size estimation techniques.	Jun-25 (BCA-204) Q5(a)	5	In syllabus
3	Explain Halstead Software Science metrics.	Jul-23 (BCA-204) Q4(a)	6	In syllabus
4	For a program with unique operators $n_1=40$, unique operands $n_2=60$, $N_1=16$ and $N_2=21$, compare: (i) Program Volume (ii) Potential Volume (iii) Program Level (iv) Program Difficulty (v) Effort (vi) Time.	Jul-23 (BCA-204) Q5(a)	6	In syllabus
5	Explain Halstead Software Science measure for any five measures.	Jun-25 (BCA-108T) Q1(f)	5	In syllabus
6	Compute the function point value for a project: user inputs = 32, user outputs = 60, user inquiries = 24, files = 08, external interfaces = 2. Assume all complexity adjustment values are average; assume 14 algorithms have been counted (as printed).	Jul-23 (BCA-204) Q4(b)	6.5	In syllabus
7	Compute the function point value: user inputs = 30, outputs = 42, enquiries = 08, files = 07, external interfaces = 6. Assume all complexity adjustment values are moderate. What are the unadjusted and adjusted function point counts?	Jun-25 (BCA-204) Q5(b)	5	In syllabus
8	Project with 15 low external inputs, 10 high external outputs, 10 low internal logical files, 15 high external interface files, 10 average external enquiries; also essential backup & recovery, significant data communication, designed code moderately reusable; other adjustment factors average. Find unadjusted and adjusted function points (weighting table: EI 3/4/6, EO 4/5/7, EQ 3/4/6, ILF 7/10/15, EIF 5/7/10 for Low/Avg/High).	Jun-25 (BCA-108T) Q4(a)	5	In syllabus

Cost Estimation (COCOMO)

#	Question	Source	Marks	Status
1	What is the basic COCOMO model?	Jun-25 (BCA-204) Q1(c)	4	In syllabus
2	Explain different models of COCOMO Model.	Jul-23 (BCA-204) Q5(b)	6.5	In syllabus
3	Explain Cost Estimation Model COCOMO in brief.	Jun-25 (BCA-108T) Q4(b)	5	In syllabus

#	Question	Source	Marks	Status
4	Discuss the COCOMO model in detail. For a 600 KLOC project, calculate effort and development time for organic, semi-detached and embedded modes.	Jun-24 (BCA-204) Q4	12.5	In syllabus
5	For a 400 KLOC project, calculate effort and development time for organic, semi-detached and embedded modes (coefficient table a,b,c,d given: Embedded 2.4/1.05/2.5/0.38, Semi-detached 3.0/1.12/2.5/0.35, Organic 3.6/1.20/2.5/0.32).	Jun-25 (BCA-204) Q4(b)	5	In syllabus
6	Compare the Walston-Felix and SEL models for software involving 8 person-years effort: (i) LOC produced (ii) duration (iii) productivity in LOC/PY (iv) average manning. <i>Note: Syllabus covers only COCOMO (basic & intermediate) for cost estimation; Walston-Felix and SEL models are not listed.</i>	Jun-24 (BCA-204) Q5	12.5	NOT IN SYLLABUS
7	Explain the Walston & Felix model and compare with the SEL model. <i>Note: Walston-Felix / SEL models are not in the syllabus.</i>	Jun-25 (BCA-204) Q4(a)	5	NOT IN SYLLABUS

Risk Management

#	Question	Source	Marks	Status
	No question on Risk Management appeared in any of the four papers.			

UNIT III

Syllabus: Design: cohesion & coupling and their classification. Quality management: quality concept, SQA, TQM, review, inspection. Implementation: structured coding, coding style, standards & guidelines, documentation guidelines, reverse engineering, re-engineering, configuration management.

Software Design (Cohesion & Coupling)

#	Question	Source	Marks	Status
1	Explain different types of coupling.	Jul-23 (BCA-204) Q6(a)	6	In syllabus
2	Explain different types of Cohesion.	Jul-23 (BCA-204) Q7(a)	6	In syllabus
3	What is a software module and cohesion? Discuss the various types of module cohesions.	Jun-24 (BCA-204) Q6	12.5	In syllabus
4	Define module coupling and explain different types of coupling.	Jun-25 (BCA-204) Q7(a)	5	In syllabus
5	Define coupling and explain various types of coupling. Which one is best and why?	Jun-25 (BCA-108T) Q7(a)	5	In syllabus
6	What is design? Describe the difference between conceptual design and technical design. <i>Note: Under Software Design the syllabus lists only cohesion & coupling.</i>	Jun-25 (BCA-204) Q6(b)	5	NOT IN SYLLABUS
7	Define: Objects, Message, Abstraction, Class, Inheritance and Polymorphism. <i>Note: Object-oriented concepts are not in the syllabus.</i>	Jun-25 (BCA-204) Q7(b)	5	NOT IN SYLLABUS

Quality Management

#	Question	Source	Marks	Status
1	Explain software quality assurance and its activities.	Jul-23 (BCA-204) Q6(b)	6.5	In syllabus
2	What is software review and software inspection?	Jul-23 (BCA-204) Q1(d)	5	In syllabus
3	Short note: Total Quality Management (TQM). [Q7 has three short notes for 12.5 marks in all]	Jun-24 (BCA-204) Q7(a)	12.5*	In syllabus
4	What do you mean by Total Quality Management (TQM)? Explain.	Jun-25 (BCA-108T) Q1(h)	5	In syllabus
5	What are the key components of quality management in software engineering?	Jun-25 (BCA-204) Q6(a)	5	In syllabus
6	What is Quality in Software? How do we determine software quality assurance? Explain various SQA activities.	Jun-25 (BCA-108T) Q6	10	In syllabus
7	What is CMM? How is it different from ISO 9000? <i>Note: Quality concept, SQA, TQM, review and inspection are listed; CMM and ISO 9000 are not.</i>	Jun-25 (BCA-108T) Q7(b)	5	NOT IN SYLLABUS

Implementation, Re-engineering & Configuration Management

#	Question	Source	Marks	Status
1	What is software re-engineering?	Jul-23 (BCA-204) Q1(b)	5	In syllabus
2	What is software reverse engineering?	Jun-24 (BCA-204) Q1(d)	5	In syllabus
3	What do you mean by Reverse Engineering? Explain.	Jun-25 (BCA-108T) Q1(b)	5	In syllabus
4	What is re-engineering? Differentiate between re-engineering and new development.	Jun-25 (BCA-204) Q8(b)	5	In syllabus
5	Short note: Software Re-Engineering. [Q7(c), shared 12.5 marks]	Jun-24 (BCA-204) Q7(c)	12.5*	In syllabus

#	Question	Source	Marks	Status
6	What do you understand by Configuration Management?	Jul-23 (BCA-204) Q7(b)	6.5	In syllabus
7	Short note: Configuration Management. [Q7(b), shared 12.5 marks]	Jun-24 (BCA-204) Q7(b)	12.5*	In syllabus

UNIT IV

Syllabus: Testing: process, levels (unit, integration, system), manual & automation, black / white / grey box, validation, verification, alpha-beta, acceptance, functional & structural testing, testing vs debugging. Maintenance: management, process, types (preventive, perceptive, adaptive, corrective), tools & techniques.

Software Testing

#	Question	Source	Marks	Status
1	What is software testing? Discuss the role of software testing during the software life cycle and why is it so difficult?	Jun-25 (BCA-204) Q8(a)	5	In syllabus
2	What are the steps followed in testing?	Jun-25 (BCA-204) Q1(f)	4	In syllabus
3	Define: (i) Fault (ii) Failure (iii) Bug (iv) Mistake. <i>Note: Testing terminology is not named in the syllabus; it supports 'Testing process'.</i>	Jun-25 (BCA-204) Q9(a)	6	PARTIAL
4	What is the difference between validation and verification?	Jul-23 (BCA-204) Q1(e)	5	In syllabus
5	What is verification and validation?	Jun-25 (BCA-204) Q1(g)	4	In syllabus
6	Differentiate between Verification and Validation.	Jun-25 (BCA-108T) Q1(d)	5	In syllabus
7	Differentiate between Alpha Testing and Beta Testing.	Jun-25 (BCA-108T) Q1(e)	5	In syllabus
8	Differentiate between White Box Testing and Black Box Testing with suitable examples.	Jun-25 (BCA-108T) Q9(a)	5	In syllabus
9	Differentiate between structural and functional testing.	Jun-24 (BCA-204) Q1(a)	5	In syllabus
10	Explain Functional Testing with example.	Jul-23 (BCA-204) Q8(b)	6.5	In syllabus
11	Explain Structural Testing using example.	Jul-23 (BCA-204) Q9(b)	6.5	In syllabus
12	Identify types of a triangle from three sides in the domain (1,100). Generate the total number of test cases for Equivalence Class Testing and Decision Table based testing. <i>Note: Falls under Functional Testing and its types.</i>	Jun-24 (BCA-204) Q8	12.5	In syllabus
13	Explain Path Testing with code for finding the largest of three numbers. Find cyclomatic complexity, and mention independent paths and test cases derived. <i>Note: Falls under Structural Testing.</i>	Jun-25 (BCA-108T) Q8	10	In syllabus
14	Define cyclomatic complexity. <i>Note: Not named in the syllabus; belongs to Structural Testing.</i>	Jun-24 (BCA-204) Q1(f)	5	PARTIAL

Software Maintenance

#	Question	Source	Marks	Status
1	What is software maintenance? Describe different categories of software maintenance.	Jul-23 (BCA-204) Q8(a)	6	In syllabus
2	Explain maintenance tools and its techniques.	Jul-23 (BCA-204) Q9(a)	6	In syllabus
3	Discuss the need of software maintenance and the different types of software maintenance.	Jun-24 (BCA-204) Q9	12.5	In syllabus
4	What is software maintenance? Describe various categories of maintenance. Which category consumes maximum effort and why?	Jun-25 (BCA-204) Q9(b)	4	In syllabus
5	Explain the Maintenance Process and types of maintenance in brief.	Jun-25 (BCA-108T) Q9(b)	5	In syllabus